TASK 03 – PROMPTING FOR TASK AUTOMATION

Objective

The objective of this task is to design a reusable prompt that can automatically transform unstructured information into a consistent and structured format. The prompt was tested with multiple inputs to evaluate its reliability and consistency.

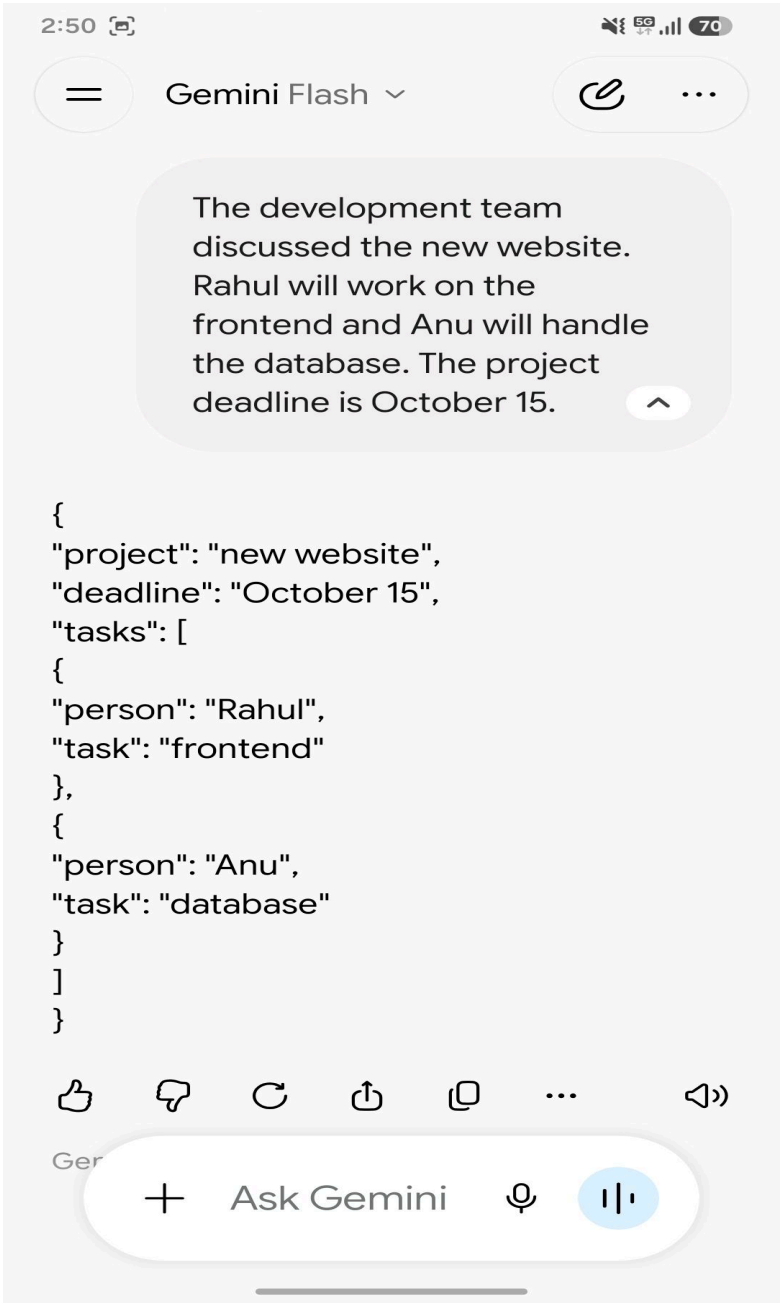
Task Selected

Converting Notes into Structured JSON

The task is to take ordinary text containing information such as people, tasks, deadlines, and project details and automatically convert it into a structured JSON format.

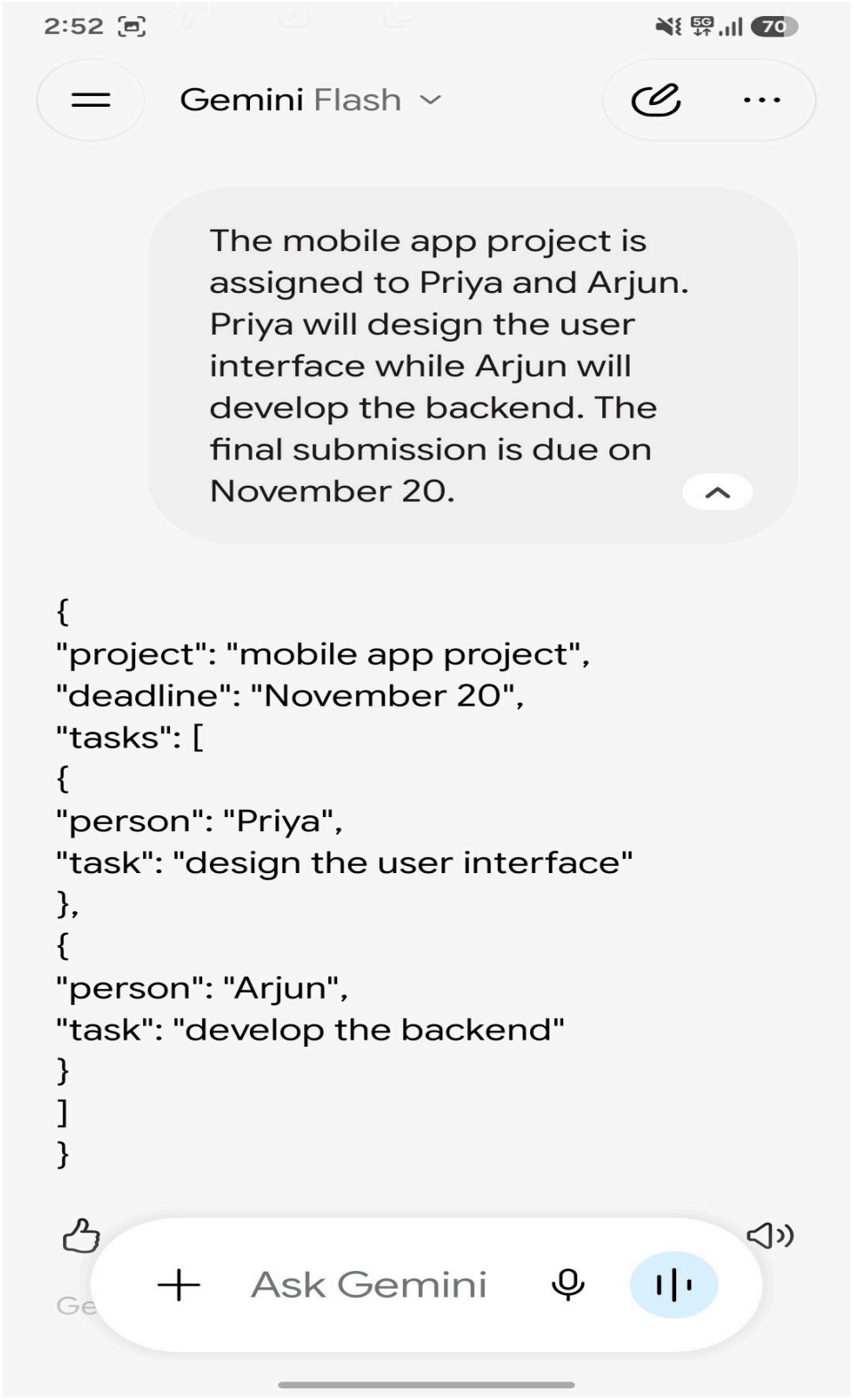
1)Input:

The development team discussed the new website. Rahul will work on the frontend and Anu will handle the database. The project deadline is October 15.



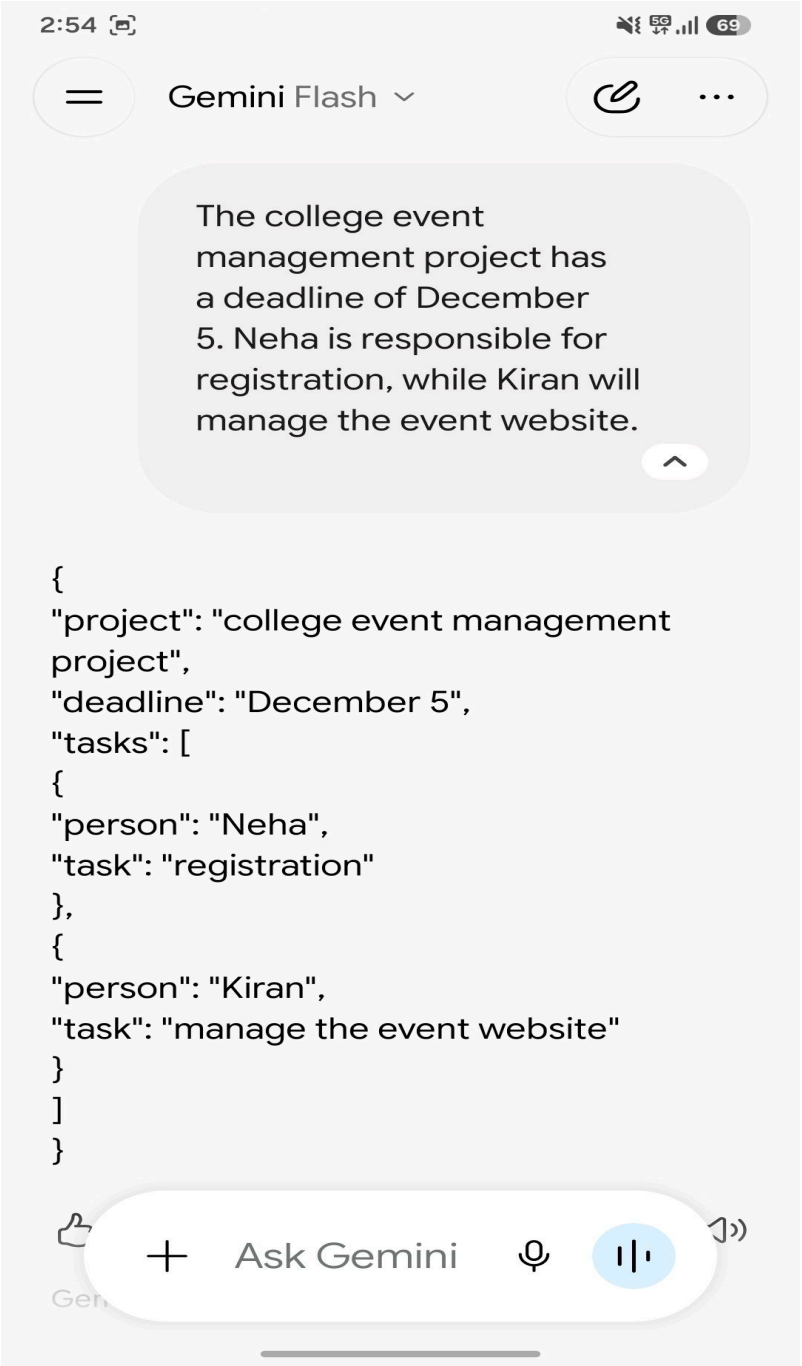
2)Input

The mobile app project is assigned to Priya and Arjun. Priya will design the user interface while Arjun will develop the backend. The final submission is due on November 20.



3)Input

The college event management project has a deadline of December 5. Neha is responsible for registration, while Kiran will manage the event website.



The prompt is reusable because the instructions and output format remain unchanged while different project notes can be provided as input. This allows the same prompt to process multiple types of project information and produce a consistent JSON structure.

Initial Version: The initial prompt asked the AI to convert project notes into JSON but did not specify an exact output structure.
Problem Identified: The responses could contain additional explanations or inconsistent field names and formatting.
Improvement: The prompt was modified to specify the exact JSON structure and instruct the AI to return only valid JSON without additional explanations.

Final Result: The improved prompt produced more consistent and structured outputs across different inputs.

The same prompt successfully processed different project descriptions and converted them into a consistent JSON structure. Providing explicit formatting instructions helped reduce variations in the output.

Key Learnings

A reusable prompt can automate repetitive information-processing tasks.
Clear output instructions improve consistency.
Structured formats such as JSON make extracted information easier to process.
Testing the prompt with different inputs helps identify inconsistencies.
Prompt iteration can improve the reliability of AI-generated outputs.

Conclusion

This task demonstrated how prompt engineering can be used to semi-automate a real-world data extraction task. By designing a reusable prompt with clear instructions and a fixed output structure, different unstructured inputs could be converted into consistent JSON outputs.